

Rationale

As a result of my planning of the time scales, tasks and costs of the project, I believe that although it is possible to complete the project in the requested time frame of 17 weeks, the timeline will still be very tight and would only make the deadline by a couple of days.

Additionally, the project would be affordable and continue to return good levels of profit over the next three years.

In this report I will highlight some of the key decisions I made and how these influenced my final findings. I will also provide a rationale for my decisions and cover potential impacts on the project.

Order of tasks considering dependencies and prerequisites

One of the first set of tasks I decided to schedule was the upgrading of the infrastructure and installing the physical server. I scheduled this first as I think it is vital that we get the target hardware completed so that modules can be deployed and tested on the systems that they will be used on as soon as possible. Also this means that the Network Engineer (Marius) can be freed up to work on other projects for the company as soon as this task is completed.

The first module I scheduled for development was the back-end database, many of the other modules are likely to either draw data from this, or need to write data to it, so it is important that this is completed early in the project.

In order to ensure I met the very tight timelines, I did have to schedule one module to be developed at the same time as the back-end database, specifically the investments and savings module. While this is likely to need part of the database, it would probably also need external data, so it would be possible to complete some of the functionality before the database is complete, and as more of the database is developed they could integrate that functionality. I scheduled integration testing for these two modules to happen at the same time, so that Gina and Sarah could work closely to ensure that the two modules share data as needed and any issues are resolved before the rest of the project continues.

Because of the difficulty of developing these two modules at the same time, I anticipate that one of the project's three major faults could happen during this phase, therefore I have added this to my plan. For the other predicted faults I have chosen to spread these out across the whole project, as it is difficult to predict exactly when they will occur. However, I have scheduled them after integration of modules happens, as this is firstly a possible time when issues could occur, but also allows me to anticipate possible delays at different stages of the project, and therefore be able to stick roughly to the subsequent times if an issue does occur.

For each of the modules I chose to take an agile approach. There are three predicted minor faults for each module, which typically would be revealed through testing, so I have chosen to split each module into three “sprints” allowing 7 hours of testing at each stage (meeting the total for each module of 21 hours of testing).

Allocation of tasks to staff

I have allocated staff based on their skill and experience, as provided in the task information, and I have tried to ensure that the chosen modules are appropriate for the staff member. As such Sarah (the Senior Developer) was allocated to the Investment and savings module. Because this is both important, and as I have mentioned earlier is a difficult task to complete while the database is still also being developed, I thought it would be sensible to allocate the most experienced software engineer to this task.

Gina is a Data specialist, so I have mainly only used her for the back-end database module, however I have also allocated her to work on the fixing and regression testing of any major faults along with Sarah. I have allocated both staff members to this as the system is heavily reliant on data so a major fault is likely to involve a number of software and data components, therefore using the most experienced Software Engineer and a data specialist should allow us to tackle any problems effectively.

The junior developers I have always allocated to work together on the same task. I have chosen to do this because Ahad is very inexperienced having only just graduated university, so I would not expect him to work independently on modules especially on such a high-risk system.

Potential benefits and risks

One particular concern that I noticed was that there is no testing task scheduled for hardware testing. I would assume that the engineer should test the infrastructure and server as they go, but not having it specifically scheduled could lead to specific testing not being carried out resulting in issues later.

Choosing to produce the investments module at the same time as the database is very risky because if part of the database that is needed for the investments module isn't completed early in the process, this would delay the whole project. However, the two staff I have allocated are very experienced and as such should be aware of such potential issues, which can be prevented or planned for through communication between the two staff members.

The agile method I have chosen is a definite benefit, by breaking the development of each module into shorter sprints, and pairing these with testing and fixing of minor faults, the impact of a fault should be lessened compared to doing all the testing in one go. As such this should result in fewer unforeseen issues arising and allow us to stick to the predicted timings.

I chose to use a cloud server for the company, although it is more expensive over the course of three years, the initial set up cost is lower, and it would provide better scalability for a growing business.

I did notice that there was no time allocation predicted for the cloud server, I assume that because it doesn't require any physical hardware and they can be set up quite quickly that this would form part of the Module development and deployment tasks. However, I think this is very risky indeed and there should be some specific times allocated in the plan for it. I have therefore allocated the same time as the physical server for Marius to work on this task.

Given the sensitive nature of the data (people's financial data) hosting on the cloud does present a security risks, so a reliable cloud provider would need to be chosen

The impact of planning decisions on the final timings and costs of the project

My planning suggests that the project would able to be completed on time. However, to achieve this I have not allocated any contingency time into the plan. As such any delay would result in the project going over deadline.

I chose the cloud server due to its reduced initial set up costs. Although this is more expensive in the long run, the trade of between scalability and the fact that the underlying system hardware would be maintained by a third party, resulting in greater reliability is worth paying extra for.

Choosing to pair the two junior software engineers on all the tasks they do, does add a little more expense to the project, but I think this is a good investment as it allows the two workers to team code and check each other's work resulting in much higher quality products.

Overall, my decisions make only minimal impact on the overall affordability of the project, the project is very affordable based on the company's current income and would result in a significant increase in profit over the next three years.